

EXPERIMENT 27: MEDICAL DIAGNOSIS

Aim

To write a **Prolog program for medical diagnosis** using symptoms and rules to identify a possible disease.

Algorithm

1. Start.
2. Define diseases and their symptoms.
3. Create rules connecting symptoms with diseases.
4. Query the symptoms or disease.
5. Match the available symptoms with the rules.
6. Display the possible diagnosis.
7. Stop.

Source Code

```
% Medical Diagnosis System
```

```
symptom(fever).
```

```
symptom(cough).
```

```
symptom(headache).
```

```
symptom(body_pain).
```

```
symptom(sore_throat).
```

```
disease(flu) :-
```

```
    symptom(fever),
```

```
    symptom(cough),
```

```
    symptom(body_pain).
```

```
disease(common_cold) :-
```

```
    symptom(cough),
```

```
symptom(sore_throat).
```

```
disease(viral_fever) :-
```

```
    symptom(fever),
```

```
    symptom(headache).
```

Query

```
?- disease(flu).
```

Output

```
?- symptom(fever).  
?- symptom(cough).  
?- symptom(headache).  
?- symptom(body_pain).  
?- symptom(sore_throat).  
?- disease(flu) :-  
    symptom(fever),  
    symptom(cough),  
    symptom(body_pain).  
?- disease(common_cold) :-  
    symptom(cough),  
    symptom(sore_throat).  
?- disease(viral_fever) :-  
    symptom(fever),  
    symptom(headache).  
  
?- disease(flu).  
true.  
  
?-
```

Result

The Prolog program successfully demonstrates a simple **rule-based medical diagnosis system** by matching symptoms with predefined disease rules.